CALCULUS · THE FUNDAMENTAL THEOREM OF CALCULUS

Accumulation Function Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Accumulation Function Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If $f(t) > 0$ on an interval, is $A(x) = \int_0^x f(t)\,dt$ increasing or decreasing there?

2. Where do you think A has a maximum?

2 Explore use the tool

1. $f(t) = 2 - t$ with $a = 0$: record $A(1)$, $A(2)$, $A(3)$ and $A(4)$.

2. Where is A largest? What is f there?

3. $f(t) = 2 \sin t$: compare the slope of A with the height of f at $x = 1.6$ and at $x = 4$.

4. Keep x the same and change a from 0 to 1. What changes and what stays the same?

3 Record use the tool and fill in the table

Accumulation values

Round to 2 decimal places.

f(t)	a	x	A(x)	f(x) = slope of A	A increasing or decreasing?
$2 - t$	0	3			
$2 \sin t$	0	2			
piecewise	0	1			
piecewise	0	3			
$2 - t$	1	3			

4 Explain what did you discover?

1. $A'(x) =$ _____.

2. A increases where f is _____ and decreases where f is _____.

3. A has a maximum where f changes from _____ to _____.

4. Changing a _____ the graph of A but does not change its _____.

5. Use the Fundamental Theorem of Calculus to explain why $\frac{d}{dx}\int_1^x t^2\,dt = x^2$.

5 Apply a new situation, no tool

1. $F(x) = \int_0^x (t^2 - 4)\,dt$. Find $F'(3)$.

2. $G(x) = \int_2^x \cos t\,dt$. Where on $[0, 2\pi]$ does G have a local maximum?

3. If $\int_0^5 f(t)\,dt = 12$ and $\int_0^2 f(t)\,dt = 4$, find $\int_2^5 f(t)\,dt$.

4. Find $\frac{d}{dx}\int_0^{x^2} \sin t\,dt$.